# Representation Theory of $\mathrm{SL}_n(\mathbb{Z})$

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**Abstract**

We give a short summary of the irreducible representations of $\mathrm{SL}_n(\mathbb{Z})$ for $n \geq 3$ and prove the super-rigidity principle for $\mathrm{SL}_n(\mathbb{Z})$. The original proof is due to Bass, Milnor, and Serre's solution to the congruence subgroup problem for $\mathrm{SL}_n(\mathbb{Z})$ [BMS67] and uses the theory of analytic pro-$p$ groups. We present the special case of that argument adapted to $\mathrm{SL}_n(\mathbb{Z})$.

## 1 Introduction

Let $\mathbf{k}$ be a field of characteristic zero and fix $n \geq 3$. Representations of $\mathrm{SL}_n(\mathbb{Z})$ are induced from two sources. Firstly, every rational representation of the algebraic group $\mathrm{SL}_n(\mathbf{k})$ restricts to a representation of $\mathrm{SL}_n(\mathbb{Z})$. Secondly, if $\pi : \mathrm{SL}_n(\mathbb{Z}) \to F$ is a surjective map into a finite group $F$ and $\rho : F \to \mathrm{GL}_r(\mathbf{k})$ is a representation of $F$ over $\mathbf{k}$, then $\rho \circ \pi : \mathrm{SL}_n(\mathbb{Z}) \to \mathrm{GL}_r(\mathbf{k})$ is a representation of $\mathrm{SL}_n(\mathbb{Z})$. By the universal property of profinite completion, representations of the second type factor through as continuous representations of the profinite completion $\widehat{\mathrm{SL}_n(\mathbb{Z})}$. And when $\mathbf{k}$ is algebraically closed, both types of representations are completely classified.

It turns out that these two sources of representations are all we need to classify the representation theory of $\mathrm{SL}_n(\mathbb{Z})$. The inclusion maps $\mathrm{SL}_n(\mathbb{Z}) \hookrightarrow \mathrm{SL}_n(\mathbf{k})$ and $\mathrm{SL}_n(\mathbb{Z}) \hookrightarrow \widehat{\mathrm{SL}_n(\mathbb{Z})}$ give an inclusion $\iota_0 : \mathrm{SL}_n(\mathbb{Z}) \hookrightarrow \mathrm{SL}_n(\mathbf{k}) \times \widehat{\mathrm{SL}_n(\mathbb{Z})}$. Define a rational representation of $\mathrm{SL}_n(\mathbf{k}) \times \widehat{\mathrm{SL}_n(\mathbb{Z})}$ to be a representation $\rho : \mathrm{SL}_n(\mathbf{k}) \times \widehat{\mathrm{SL}_n(\mathbb{Z})} \to \mathrm{GL}_r(\mathbf{k})$ such that $\rho|_{\mathrm{SL}_n(\mathbf{k})}$ is rational and $\rho|_{\widehat{\mathrm{SL}_n(\mathbb{Z})}}$ is continuous.

Putman [Putb] proves the following results which together completely classify the representation theory of $\mathrm{SL}_n(\mathbb{Z})$.

**Theorem A.** *Let $\rho : \mathrm{SL}_n(\mathbb{Z}) \to \mathrm{GL}_r(\boldsymbol{k})$ be a representation. Then $\rho$ uniquely factors through a rational representation of $\mathrm{SL}_n(\boldsymbol{k}) \times \widehat{\mathrm{SL}_n(\mathbb{Z})}$, i.e. there is a unique rational representation $\hat{\rho} : \mathrm{SL}_n(\boldsymbol{k}) \times \widehat{\mathrm{SL}_n(\mathbb{Z})} \to \mathrm{GL}_r(\boldsymbol{k})$ such that $\rho = \hat{\rho} \circ \iota_0$. Moreover, any subspace $W \subseteq \boldsymbol{k}^n$ which is an $\mathrm{SL}_n(\mathbb{Z})$-subrepresentation is also an $\mathrm{SL}_n(\boldsymbol{k}) \times \widehat{\mathrm{SL}_n(\mathbb{Z})}$-subrepresentation.*

**Theorem B.** *Representations of* $\mathrm{SL}_n(\mathbb{Z})$ *decompose into direct sums of irreducibles.*

**Theorem C.** *Suppose* $\boldsymbol{k}$ *is algebraically-closed. Then the irreducible rational representations of* $\mathrm{SL}_n(\boldsymbol{k}) \times \widehat{\mathrm{SL}_n(\mathbb{Z})}$ *are precisely those of the form* $V \otimes W$ *where* $V$ *is an irreducible rational representation of* $\mathrm{SL}_n(\boldsymbol{k})$ *and* $W$ *is a continuous representation of* $\widehat{\mathrm{SL}_n(\mathbb{Z})}$.

In order to prove Theorem A, Putman assumes a general super-rigidity result of Margulis to derive:

**Theorem D.** *Let* $\rho : \mathrm{SL}_n(\mathbb{Z}) \to \mathrm{GL}_r(\boldsymbol{k})$ *be a representation. Then there exists a finite-index subgroup* $\Gamma \leq \mathrm{SL}_n(\mathbb{Z})$ *and a rational representation* $\bar{\rho} : \mathrm{SL}_n(\boldsymbol{k}) \to \mathrm{GL}_r(\boldsymbol{k})$ *such that* $\rho|_\Gamma = \bar{\rho}|_\Gamma$.

In this document we provide an alternate proof of Theorem D by instead using the theory of analytic pro-$p$ groups, as described by Bass, Milnor, and Serre [BMS67]. We also demonstrate a different way to derive the existence half of Theorem A from Theorem D. Refer to [Putb] for self-contained proofs of Theorem B, Theorem C, and the uniqueness half of Theorem A.

# 2 Preliminaries

We assume familiarity with the theory of analytic pro-$p$ groups for $p > 2$ prime, e.g. up to Chapter 9 of [DDSSM09]. We will also take for granted the representation theory of $\mathrm{SL}_n(\mathbf{k}')$ and $\mathfrak{sl}_n(\mathbf{k}')$ for fields $\mathbf{k}'$ of characteristic zero.

To investigate the representations of $\widehat{\mathrm{SL}_n(\mathbb{Z})}$, we make use of the Congruence Subgroup Property for $\mathrm{SL}_n(\mathbb{Z})$.

**Theorem 1** (Congruence Subgroup Property (CSP))**.** *Let* $\Gamma_n(\ell) := \ker(\mathrm{SL}_n(\mathbb{Z}) \to \mathrm{SL}_n(\mathbb{Z}/\ell\mathbb{Z}))$. *Given a finite index subgroup* $K$ *of* $\mathrm{SL}_n(\mathbb{Z})$, *there exists* $m \in \mathbb{N}$ *such that* $\Gamma_n(m) \leq K$.

*Proof.* We refer the reader to Putman's proof. [Puta] □

This implies that

$$\widehat{\mathrm{SL}_n(\mathbb{Z})} \cong \varprojlim_m \mathrm{SL}_n(\mathbb{Z}/m\mathbb{Z}) \cong \mathrm{SL}_n(\hat{\mathbb{Z}}).$$

By the Chinese Remainder Theorem, we also obtain that

$$\widehat{\mathrm{SL}_n(\mathbb{Z})} \cong \prod_p \mathrm{SL}_n(\mathbb{Z}_p)$$

as topological groups, where the product is taken over primes $p$.

We will also need some facts about $\mathrm{SL}_n(\mathbb{Z}_p)$ and $\mathrm{GL}_r(\mathbb{Z}_p)$ for primes $p$. These are $p$-adic analytic groups, meaning that any of their open subgroups contain

a further open uniform pro-$p$ subgroup; in fact, these uniform pro-$p$ subgroups can be taken to be one of the principal congruence subgroups.

Under the equivalence of categories between local $p$-adic analytic groups and $\mathbb{Q}_p$-Lie algebras, $\mathrm{SL}_n(\mathbb{Z}_p)$ corresponds to $\mathfrak{sl}_n(\mathbb{Q}_p)$ and $\mathrm{GL}_r(\mathbb{Z}_p)$ corresponds to $\mathfrak{gl}_r(\mathbb{Q}_p)$.

# 3 Super-rigidity proof

In this section, we will prove Theorem D. We will need the following lemma.

**Lemma 1.** *Let $R$ an integral domain of characteristic 0 which is finitely generated as a $\mathbb{Z}$-module. Then there are infinitely many primes $p \in \mathbb{Z}$ such there exists a ring embedding $R \hookrightarrow \mathbb{Z}_p$.*

*Proof.* This follows from Cassels' embedding theorem [Cas76]. Let $r_1, ..., r_m$ be a set of nonzero generators for $R$ and let $F$ be the field of fractions of $R$. By Cassels' embedding theorem, there are infinitely many primes $p$ for which we have an embedding $\alpha : F \to \mathbb{Q}_p$ such that $|\alpha(r_i)| \leq 1$ for all $i$. Then clearly $\alpha|_R$ is a ring embedding that maps into $\mathbb{Z}_p$. □

*Proof of Theorem D.* $\mathrm{SL}_n(\mathbb{Z})$ is finitely-generated, so let $A_1, ..., A_m$ be generators. $\rho$ is determined by its values on the $A_i$, so if $R$ is the subring of $\mathbf{k}$ generated by $\mathbb{Z}$ as well as the matrix entries of $\rho(A_i)$ for all $i$, then $\rho$ maps into $\mathrm{GL}_r(R)$. By Lemma 1, pick a prime $p > 2$ such that $R$ embeds in $\mathbb{Z}_p$, so we can view $\rho$ as a map $\rho : \mathrm{SL}_n(\mathbb{Z}) \to \mathrm{GL}_r(\mathbb{Z}_p)$.

Since the codomain is a profinite group, this map factors through a continuous homomorphism

$$\rho' : \widehat{\mathrm{SL}_n(\mathbb{Z})} \to \mathrm{GL}_r(\mathbb{Z}_p)$$

such that the below diagram commutes.

$$\begin{array}{ccc} \mathrm{SL}_n(\mathbb{Z}) & \xrightarrow{\rho} & \mathrm{GL}_r(\mathbb{Z}) \\ \downarrow{\scriptstyle \iota} & \nearrow_{\rho'} & \\ \widehat{\mathrm{SL}_n(\mathbb{Z})} & & \end{array}$$

Recall that

$$\widehat{\mathrm{SL}_n(\mathbb{Z})} \cong \prod_q \mathrm{SL}_n(\mathbb{Z}_q).$$

Let $\Gamma_1$ be the kernel of the surjection $\mathrm{GL}_r(\mathbb{Z}_p) \to \mathrm{GL}_r(\mathbb{Z}/p\mathbb{Z})$, which is an open uniform pro-$p$ subgroup of $\mathrm{GL}_r(\mathbb{Z}_p)$. Let $\Gamma' = (\rho')^{-1}(\Gamma_1) \subseteq \prod_q \mathrm{SL}_n(\mathbb{Z}_q)$, which is an open and finite-index subgroup. By the product topology, there is an open subset $\Gamma''$ of $\prod_q \mathrm{SL}_n(\mathbb{Z}_q)$ such that

$$\Gamma'' := \prod_q U_q \subseteq \Gamma'$$

where $U_q$ is an open compact subgroup of $\mathrm{SL}_n(\mathbb{Z}_q)$ and $U_q = \mathrm{SL}_n(\mathbb{Z}_q)$ for all but finitely many $q$. As $\mathrm{SL}_n(\mathbb{Z}_p)$ is $p$-adic analytic, by shrinking $U_p$ if necessary, we can assume that $U_p$ is uniform. (For instance, by using the congruence subgroup property for $\mathrm{SL}_n(\mathbb{Z}_p)$, we can take $U_p = \ker(\mathrm{SL}_n(\mathbb{Z}_p) \to \mathrm{SL}_n(\mathbb{Z}_p/p^k\mathbb{Z}_p))$ for some large $k$.

**Lemma 2.** *$\rho'|_{U_q} : U_q \to \Gamma_1$ has trivial image for every $q \neq p$.*

*Proof.* We first show the image of $\rho'$ is finite. Since $U_q$ is an open subgroup of $\mathrm{SL}_n(\mathbb{Z}_q)$, it is a $q$-adic analytic group. Let $K \leq U_q$ be some uniform pro-$q$ open subgroup. Then $\rho'(K) \subseteq \Gamma_1$ is a pro-$q$ group, being the continuous image (and hence a quotient by a closed subgroup) of $K$; and is a closed (compact) subgroup of the pro-$p$ group $\Gamma_1$. Hence $\rho'(K)$ is also pro-$p$. So $\rho'(K) = \{1\}$. Now, since $K$ has finite index in $U_q$, we can cover $U_q$ with finitely many translations of $K$. Thus, the image $\rho'(U_q)$ is a finite subgroup of $\Gamma_1$.

We are now done, because $\Gamma_1$ is uniform and thus torsion-free. Hence, it has no non-trivial finite subgroups. □

Let $\pi_p : \Gamma'' \to U_p$ be the projection map and $\tilde{\rho} := \rho'|_{U_p}$. Then $\rho' = \tilde{\rho} \circ \pi_p$ by a pointwise convergence argument, by Lemma 2 and because $\rho'$ is continuous.

Let $\Gamma = \iota^{-1}(\Gamma'')$ which is a finite-index subgroup of $\mathrm{SL}_n(\mathbb{Z})$. Then we have shown that $\rho|_\Gamma : \Gamma \to \Gamma_1$ extends to a map $\tilde{\rho} : U_p \to \Gamma_1$, i.e. that the following diagram commutes.

$$\begin{array}{ccc} \Gamma & \xrightarrow{\rho|_\Gamma} & \Gamma_1 \\ \downarrow{\scriptstyle \iota} & & \uparrow{\scriptstyle \tilde{\rho}} \\ \prod_q U_q & \xrightarrow{\pi_p} & U_p \end{array}$$

Recall that $\Gamma_1$ and $U_p$ are both uniform pro-$p$ groups. In the language of Lazard's theory [Laz65], $\tilde{\rho}$ represents a germ of $\mathcal{H}(\mathrm{SL}_n(\mathbb{Z}_p), \Gamma_1)$, the set of all analytic homomrphisms whose domain is an open subgroup of $\mathrm{SL}_n(\mathbb{Z}_p)$ and whose codomain is $\Gamma_1$. Via the equalivalence of categories between local $p$-adic analytic groups and finite dimensional Lie algebras over $\mathbb{Q}_p$, we obtain a corresponding Lie algebra representation over $\mathbb{Q}_p$:

$$(\tilde{\rho})^* : \mathfrak{sl}_n(\mathbb{Q}_p) \to \mathfrak{gl}_r(\mathbb{Q}_p).$$

Now $\mathbb{Q}_p$ has characteristic 0, so such representations are completely classified: they decompose into direct sums of irreducible representations, and irreducible representations are given by applying Schur functors to the standard representation. So our Lie algebra representation $(\tilde{\rho})^*$ can be written as

$$\bigoplus_{\lambda_i} \mathbb{S}_{\lambda_i}(\mathbb{Q}_p^n) \tag{1}$$

for some collection of Young tableaux $\lambda_i$.

However, the same expression in (1) defines an $\mathrm{SL}_n(\mathbb{Q}_p)$-representation $\varphi : \mathrm{SL}_n(\mathbb{Q}_p) \to \mathrm{GL}(V)$, whose induced Lie algebra representation is also $(\tilde{\rho})^*$. So

by [DDSSM09] Theorem 9.11, we have that $\tilde{\rho} : U_p \to \Gamma_1$ and $\varphi$ agree on any uniform open subgroup around $1 \in \mathrm{SL}_n(\mathbb{Q}_p)$ on which they are both defined; in particular, they agree on $U_p$. So $\varphi$ is an extension of $\tilde{\rho}$, and $\varphi$ is a rational map of algebraic groups where the polynomials are given by integer coefficients. Hence $\tilde{\rho}$ is also rational with integer coefficients.

Now, the inclusion and projection $\Gamma \xrightarrow{\iota} \Gamma'' \xrightarrow{\pi_p} U_p$ are also rational with integer coefficients, hence our original representation $\rho|_\Gamma : \Gamma \to \Gamma_1$ is rational with integer coefficients. But recall that the codomain of $\rho$ can in fact be taken to be $\mathrm{GL}_r(\mathbf{k})$. And polynomials with integer coefficients can take inputs and outputs in $\mathbf{k}$. Hence, using the integer cofficient polynomials defining $\rho|_\Gamma$, we can extend to a rational map $\bar{\rho} : \mathrm{SL}_n(\mathbf{k}) \to \mathrm{GL}_r(\mathbf{k})$, where $\rho|_\Gamma = \bar{\rho}|_\Gamma$.

□

We have now proven Theorem D. Putman's [Putb] proof of Theorem A suffices to conclude here, but we present here an alternative argument for the existence half of Theorem A.

**Lemma 3.** *Let $G$ be a group and $H \leq G$ be a finite-index subgroup. Let $\rho : G \to \mathrm{GL}(V)$ be a representation. Then $\rho$ is a quotient of $\mathrm{Ind}_H^G \mathrm{Res}_H^G(\rho)$.*

*Proof.* Let $W = \mathrm{Res}_H^G \rho$. By adjunction,

$$\mathrm{Hom}_G(\mathrm{Ind}_H^G W, V) \cong \mathrm{Hom}_H(W, W) \neq 0.$$

In particular, there exists a surjective map $f \in \mathrm{Hom}_H(W, W)$. Left adjoints preserve epimorphisms, hence there is a corresponding surjective map $\mathrm{Ind}_H^G f \in \mathrm{Hom}_G(\mathrm{Ind}_H^G W, V)$. (Alternatively, one can take $f$ to be the identity in $W$ and calculate $\mathrm{Ind}_H^G f$ explicitly under Hom-tensor adjunction to see that it is surjective.) Hence $\rho$ is a quotient of $\mathrm{Ind}_H^G \mathrm{Res}_H^G(\rho)$. □

**Theorem 2.** *Keep the same notations as in Theorem D. Then $\rho$ factors through a rational representation $\hat{\rho} : \widehat{\mathrm{SL}_n(\mathbb{Z})} \times \mathrm{SL}_n(\boldsymbol{k}) \to \mathrm{GL}_r(\boldsymbol{k})$.*

*Proof.* Let $\psi = \mathrm{Ind}_\Gamma^{\mathrm{SL}_n(\mathbb{Z})} \rho|_\Gamma$. By Lemma 3, $\rho(g) = \pi \circ \psi(g)$ for all $g \in \mathrm{SL}_n(\mathbb{Z})$ for some surjective representation morphism $\pi$. It thus suffices to prove the result for $\psi$, i.e. that $\psi$ factors through a rational representation $\hat{\psi}$ of $\widehat{\mathrm{SL}_n(\mathbb{Z})} \times \mathrm{SL}_n(\mathbf{k})$, and then setting $\hat{\rho}(g) = \pi \circ \hat{\psi}(g)$.

Let $t_1, ..., t_d$ be a system of coset representatives of $\mathrm{SL}_n(\mathbb{Z})/\Gamma$. For $g \in \mathrm{SL}_n(\mathbb{Z})$, let $j_g : \{1, ..., d\} \to \{1, ..., d\}$ be the permutation such that for all $i$, $j_g(i)$ is the unique number with $t_{j_g(i)}^{-1} g t_i \in \Gamma$. Then $\psi(g)$ is the block-permutation matrix whose $ij$ block is zero if $j \neq j_g(i)$ and equal to $\rho(t_{j_g(i)}^{-1} g t_i)$ otherwise.

Let $C(\Gamma) = \bigcap_{g \in \mathrm{SL}_n(\mathbb{Z})} g\Gamma g^{-1}$ be the normal core of $\Gamma$. Notice that $j_g$ depends only on the image of $g$ in $\mathrm{SL}_n(\mathbb{Z})/C(\Gamma)$, because if $h \in C(\Gamma)$ then $t_{j_g(i)}^{-1} g h t_i = t_{j_g(i)}^{-1} g t_i t_i^{-1} h t_i \in \Gamma$.

We will now define a representation $\hat{\psi}$ of $\mathrm{SL}_n(\mathbb{Z})/C(\Gamma) \times \mathrm{SL}_n(\mathbf{k})$, which automatically factors through a continuous representation of $\widehat{\mathrm{SL}_n(\mathbb{Z})} \times \mathrm{SL}_n(\mathbf{k})$.

Given $(gC(\Gamma), A) \in \mathrm{SL}_n(\mathbb{Z})/C(\Gamma) \times \mathrm{SL}_n(\mathbf{k})$, define $\hat{\psi}(gC(\Gamma), A)$ to be the permutation matrix whose nonzero blocks are $\bar{\rho}(t_{j_g(i)}^{-1} A t_i)$, where $\bar{\rho}$ is as in Theorem D. We show that $\hat{\psi}$ is a group homomorphism.

We first show $g \mapsto j_g$ is a homomorphism. This is true because

$$t_{j_g(j_h(i))}^{-1} ght_i = t_{j_g(j_h(i))}^{-1} g t_{j_h(i)} t_{j_h(i)}^{-1} ht_i \in \Gamma$$

and hence $j_{gh}(i) = j_g(j_h(i))$.

Let $E_{ij}$ be the block matrix whose $ij$-block is the $n \times n$ identity matrix and whose other blocks are 0. Let $\otimes$ denote the matrix Kronecker product. Compute

$$\begin{aligned}
&\hat{\psi}(gC(\Gamma), A)\hat{\psi}(hC(\Gamma), B) \\
=&\big(\sum_i E_{j_g(i)i} \otimes \bar{\rho}(t_{j_g(i)}^{-1} A t_i)\big)\big(\sum_l E_{j_h(l)l} \otimes \bar{\rho}(t_{j_h(l)}^{-1} B t_l)\big) \\
=&\sum_{i,l} (E_{j_g(i)i} \otimes \bar{\rho}(t_{j_g(i)}^{-1} A t_i))(E_{j_h(l)l} \otimes \bar{\rho}(t_{j_h(l)}^{-1} B t_l)) \\
=&\sum_i E_{j_g(i) j_h^{-1}(i)} \otimes \bar{\rho}(t_{j_g(i)}^{-1} AB t_{j_h^{-1}(i)}) \\
=&\hat{\psi}(ghC(\Gamma), AB).
\end{aligned}$$

Therefore $\hat{\psi}$ is a homomorphism. Also

$$\hat{\psi}|_{\mathrm{SL}_n(\mathbf{k})}(C(\Gamma), A) = \sum_i E_{ii} \otimes \bar{\rho}(t_i^{-1} A t_i)$$

which is rational because $\bar{\rho}$ is rational. And $\hat{\psi}$ clearly extends $\psi$. Therefore we are done.

□

# References


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